

Consul: This is a service discovery and configuration tool that helps the client to have an updated view of the infrastructure. The consul agent is a main process in consul. It keeps information about all nodes that are a part of the consul and runs update checks regularly. The consul agent can run in either server mode or client mode.

Docker Machine: Docker Machine is a tool that allows us to install and run Docker on virtual hosts (virtual machines). We can manage these created hosts as we manage virtual machines using the Docker Machine commands or also from a VirtualBox interface. As Docker Machine uses VirtualBox drivers, the commands of the former will not run inside a VirtualBox. It requires the host system to run the Docker Machine commands.

Let's now create an overlay network. The four steps given below will help to do so.

Creating multiple virtual hosts with the key-value store

Starting with the creation of hosts, the consul container runs in the first host, which is responsible for managing the key-value store of the network properties. The first host which we create has the consul container (key-value store) running on it, so that it becomes easier to connect further containers to access this key-value store while creating them.

To create virtual hosts, install Docker Machine, which helps in creating virtual hosts using VirtualBox drivers. The command below is used for the same.

```
$ docker-machine create -d virtualbox consul-node
```

The above command creates a key-value store virtual host using VirtualBox drivers. *consul-node* is the name of the virtual host. Docker also supports Etcd and ZooKeeper key-value stores other than consul. For this command to be successful, VirtualBox should be installed in the system. Use *ssh* to log in to the created virtual host and use it. To run the consul store inside the created virtual host, use the commands given below. These commands need to be run from inside the virtual host. This creates our first virtual host with the key-value store running on port 8500.

```
$ docker-machine ssh consul-node
$ docker run -d --name consul -p "8500:8500" -h "consul"
consul agent -server -bootstrap -client "0.0.0.0"
```

In the above command, *-d* is the detach option. This runs the container in the background and prints its container ID. *--name* is used to give a name to the container, which in this case is *consul*. To publish the container's port to the host, *-p* is used. As the service runs inside the container and we need to access outside containers, we publish the 8500 port that is inside the container and map it with port 8500 of the host. *-h* gives the container hostname, which is given here as *consul*.



Figure 12: Consul node virtual host

```
$ docker-machine create -d virtualbox --swarm --swarm-master \
> --swarm-discovery="consul://$(docker-machine ip consul-node):8500" \
> --engine-opt="cluster-store=consul://$(docker-machine ip consul-node):8500" \
> --engine-opt="cluster-advertise=eth1:2376" \
> swarm-master
Running pre-create checks...
```

Figure 13: Creating a Swarm master

```
$ docker-machine create -d virtualbox --swarm \
> --swarm-discovery="consul://$(docker-machine ip consul-node):8500" \
> --engine-opt="cluster-store=consul://$(docker-machine ip consul-node):8500" \
> --engine-opt="cluster-advertise=eth1:2376" \
> swarm-node1
Running pre-create checks...
```

Figure 14: Creating a Swarm node

-server is the bootstrap and the client is '0.0.0.0'.

Creating a Swarm master

Use the commands given in Figure 13 to create a Swarm master after setting up the key-value store.

In the command, the *--swarm* option is used to configure a machine with a Swarm and *--swarm-master* is used to configure that virtual host as the swarm master. *--engine-opt* is used to include more parameters with the key=value format. Here, the engine refers to the Docker engine. Consul is used as an external key-value store here. The Swarm manager stores networking information of the Swarm mode network and *--swarm-discovery* is used to locate the Swarm manager.

Creating Swarm nodes

Next, we'll create Swarm nodes to make an overlay network of Docker containers. In this case *--swarm-master* isn't an argument, which means this is a worker node of the swarm and not a master node.

So far we have set up the Swarm master and Swarm node. In the next step, the procedure to create an overlay network is given.

Figure 15 shows us the Swarm node, the Swarm master and the Swarm manager (consul node).

Creating multiple overlay networks

We will now create two overlay networks, and check the connectivity inside and across the network. Similar to any other network, containers in the same overlay network can communicate with each other, but they cannot do so if they are in different networks.